

UW Rating Model - Executive Brief

KCD-based respiratory history and future circulatory disease prediction

Source: 260820_UW rating Model_ML_final - copy.ipynb

30.83%

Respiratory claim share

13,796

Eligible insureds - 3y

1.746

Arrhythmia RR - 4y

0.858

Best 5y ischemic AUC

Key Messages

- Respiratory diseases (J00-J99) are the largest KCD claim category and provide a reasonable analytical starting point.
- A model using Respiratory_History alone does not adequately discriminate future hypertension or ischemic heart disease.
- Arrhythmia shows RR>1 in selected horizons, but event counts are too small to treat it as a stable standalone loading factor.
- Prediction improves materially when broader morbidity, claim frequency, recency, and healthcare-utilization information is added.

Bottom line

The main signal is not a direct respiratory-history loading; future circulatory risk is segmented much better when broader historical medical information is incorporated.

Data and Cohort

Item	Result
Total KCD-classified claims	697,699
Respiratory (J00-J99) claims	215,104 (30.83%)
Historical window	2015-05-04 to 2020-05-03
Eligible insureds - 3y / 4y / 5y	13,796 / 13,566 / 13,310

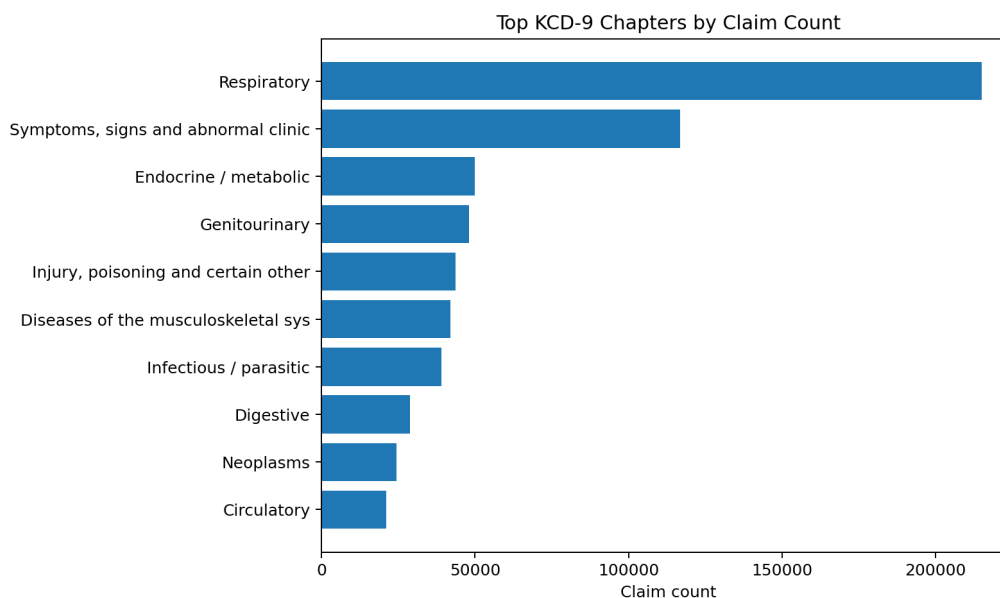


Figure 1. Top KCD-9 chapters by claim count

1. Univariate Benchmark - Respiratory History Only

The benchmark is a logistic regression using five-year Respiratory_History as the only predictor. Model-based RR shows no consistent positive loading signal for hypertension or ischemic heart disease. Arrhythmia exceeds RR=1 only at the 4-year and 5-year horizons.

Follow-up	Target	Events	AUC	Model RR
3y	Hypertension	208	0.508	0.985
3y	Ischemic Heart	53	0.471	0.820
3y	Arrhythmia	10	0.295	0.753
4y	Hypertension	262	0.480	0.817
4y	Ischemic Heart	63	0.575	0.934
4y	Arrhythmia	17	0.467	1.746
5y	Hypertension	311	0.466	1.063
5y	Ischemic Heart	75	0.539	0.666
5y	Arrhythmia	22	0.516	1.304

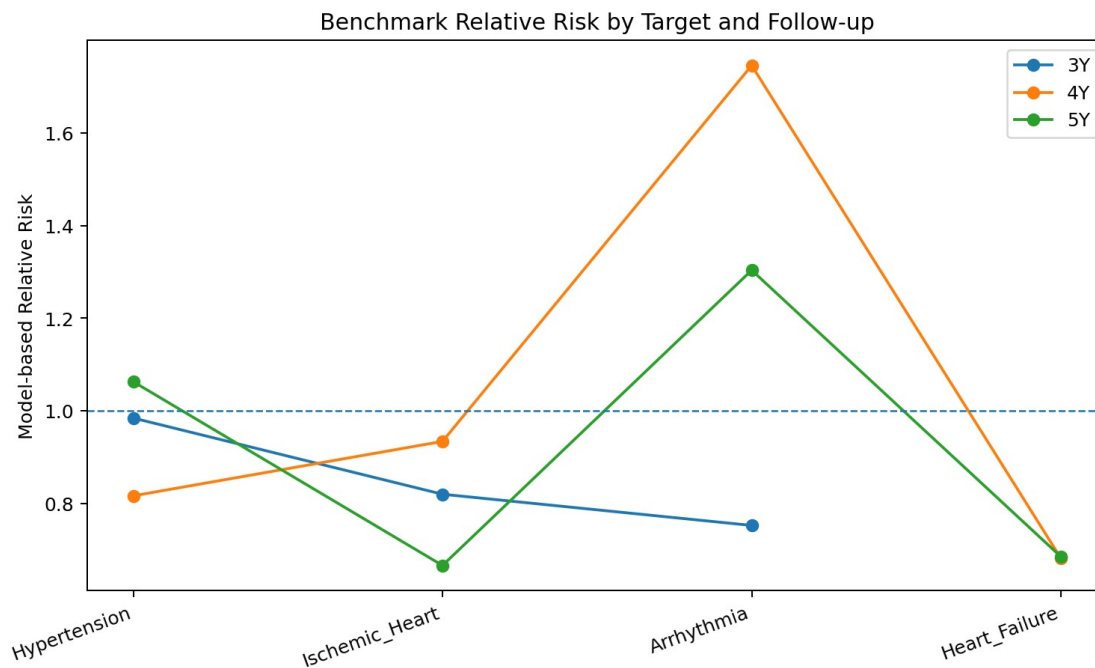


Figure 2. Model-based relative risk by benchmark target

Interpretation

RR>1 for arrhythmia is directionally notable, but the total event counts are only 17 at 4 years and 22 at 5 years. It is better treated as a signal to investigate than as a stable rating relativity.

2. Expanded Models - Broader Historical Medical Information

Expanded features include chronic respiratory history (J40-J47), respiratory claim frequency, time since the last claim, total historical claims, KCD diversity, and prior circulatory, diabetes, and metabolic history. Candidate models are compared on the same train/test split.

Follow-up	Target	Best ROC-AUC model	ROC-AUC
3y	Hypertension	Logistic Multivariate	0.809
4y	Hypertension	Random Forest	0.745
5y	Hypertension	Random Forest	0.745
3y	Ischemic Heart	Random Forest	0.790
4y	Ischemic Heart	HistGradientBoosting	0.802
5y	Ischemic Heart	Logistic Multivariate	0.858

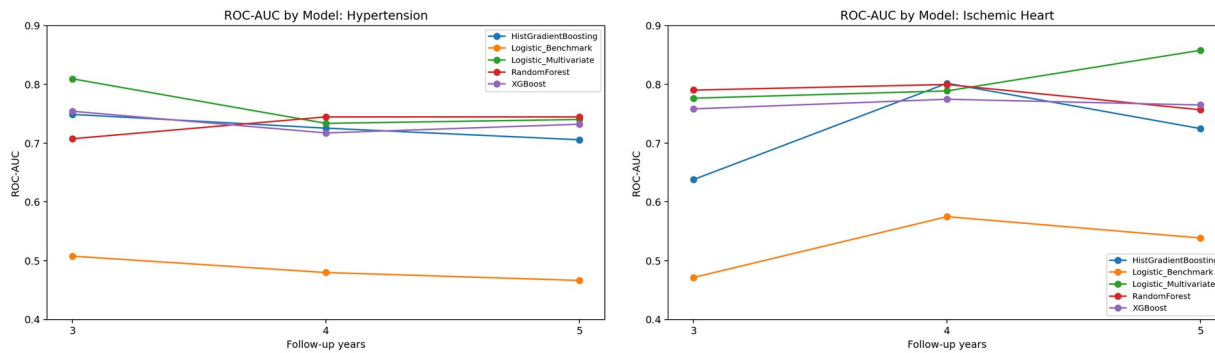


Figure 3. ROC-AUC by model: hypertension (left) and ischemic heart disease (right)

Model comparison

Expanded models materially outperform the respiratory-history-only benchmark for both targets. The strongest result is the 5-year ischemic heart disease Logistic Multivariate model with ROC-AUC 0.858.

Final Summary

- Respiratory history is meaningful as a starting feature, but its standalone predictive value is limited.
- Future circulatory risk is better segmented using broader historical morbidity and healthcare-utilization patterns.
- The most consistent performance gains are observed for hypertension and ischemic heart disease.